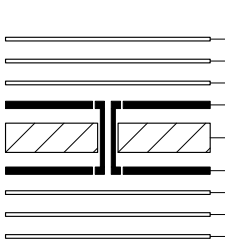
JONES_MIXER Fabrication Document

Layer Stack Legend



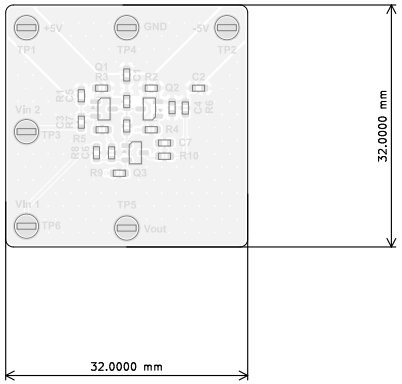
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	L1	0.035mm (1.00oz)		Signal	GBR
Core		1.471mm	FR4_7628	Dielectric	
Copper	L2	0.035mm (1.00oz)		Signal	GBR
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.581mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY LTU.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	32.000 × 32.000 mm
BOARD THICKNESS	1.581 mm
TRACE WIDTH	0.300 mm
TRACE TO TRACE	0.300 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

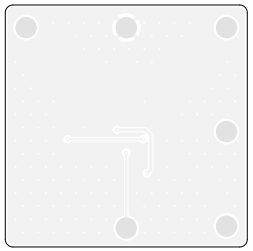
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Azalea Vega	Date: 2024-04-03	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 7

JONES_MIXER Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

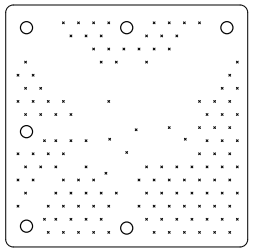
Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Azalea Vega	Date: 2024-04-03	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 7

JONES_MIXER Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	142	0.25mm (9.84mils)	PTH	Round	L1 - L2	Via
○	6	1.60mm (62.99mils)	PTH	Round	L1 - L2	Pad
	Total 148					

Drill Drawing L1 - L2 (Scale 1:1)



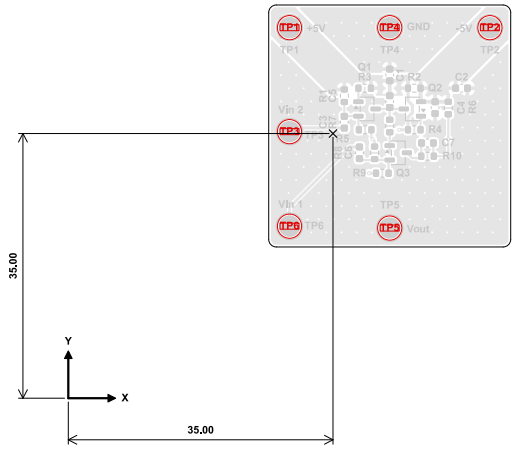
Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Drill Drawing (L1 - L2)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Andrew Vega	Date: 2024-04-03	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 7

JONES_MIXER Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	+5V	2.75	29.00
TP2	-5V	29.25	29.00
TP3	Vi2	2.75	15.25
TP4	GND	16.00	29.00
TP5	Vo	16.00	2.50
TP6	Vi1	2.75	2.75

Ref.	Net	X [mm]	Y [mm]



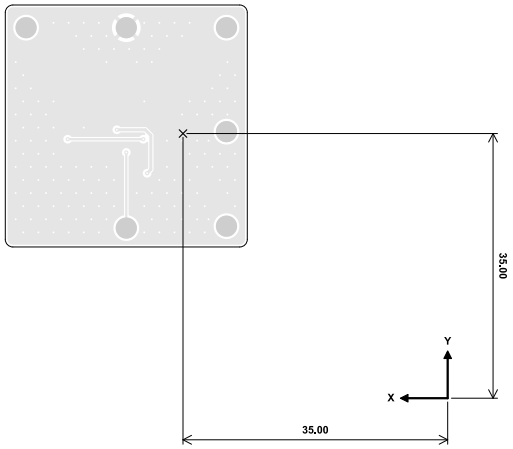
All dimensions are in millimeters unless otherwise specified.

blue devil MOTORSPORTS	Comments:	Company: LTU	Variant: CHECKED	Git Hash: 9075b89
		Board Name: JONES_MIXER	Project Name: Jones Mixer	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Azalea Vega	Date: 2024-04-13 Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4 Sheet: 4 of 7

JONES_MIXER Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

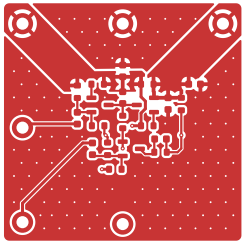


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Azalea Vega	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 7

JONES_MIXER Fabrication Document

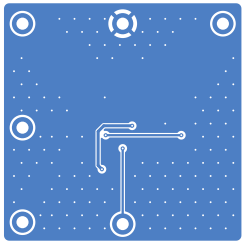
L1 (Scale 1:1)



Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: L1 (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Andrea Vega	Date: 2024-04-03	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 7

JONES_MIXER Fabrication Document

L2 (Scale 1:1)



Comments:	Company: LTU		Variant: CHECKED	Git Hash: 9075b89
	Board Name: JONES_MIXER		Project Name: Jones Mixer	
Sheet Title: L2 (Scale 1:1)	File Name: jones_mixer.kicad_pcb	Designer: Aidan Brzezinski, Chloe LaBelle, Andrew Vega	Date: 2024-04-03	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 7